

First-Order ODEs (Zill 2.2–2.5)

Separable: $\frac{dy}{dx} = g(x)h(y)$, so $\int \frac{dy}{h(y)} = \int g(x) dx + C$ – separate variables, then integrate both sides.

Linear & integrating factor: $\frac{dy}{dx} + P(x)y = Q(x)$, $\mu(x) = e^{\int P(x) dx}$, $y = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x) dx + C \right]$ – multiplying by μ makes the LHS $\frac{d}{dx}[\mu y]$.

Exact equations: $M(x, y) dx + N(x, y) dy = 0$ is exact iff $M_y = N_x$; solution $f(x, y) = \int M dx + g(y) = C$ with $f_y = N$ – potential function f has $f_x = M$, $f_y = N$.

Integrating factors for non-exact equations: $\mu(x) = e^{\int (M_y - N_x)/N dx}$ if that ratio depends on x only; $\mu(y) = e^{\int (N_x - M_y)/M dy}$ if it depends on y only – multiplying by μ restores exactness; try the x -only case first.

Homogeneous-coefficient DE (sub $y = vx$): $\frac{dy}{dx} = f(y/x)$, $y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$ – used when M, N are homogeneous of the same degree; reduces to a separable equation in v, x .

Bernoulli equation: $\frac{dy}{dx} + P(x)y = Q(x)y^n$, $n \neq 0, 1$, sub $w = y^{1-n} \Rightarrow \frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x)$ – turns it into a first-order linear equation in w .

Linear Higher-Order ODEs (Zill 3.1–3.6)

Reduction of order: given one solution y_1 of $y'' + Py' + Qy = 0$, $y_2 = y_1(x) \int \frac{e^{-\int P dx}}{[y_1(x)]^2} dx$ – produces a second, linearly independent solution $y_2 = y_1 v(x)$.

Homogeneous, constant coefficients: $ay'' + by' + cy = 0 \Rightarrow am^2 + bm + c = 0$ (sub $y = e^{mx}$ gives the auxiliary equation). Three root cases:

Distinct real m_1, m_2 : $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$.

Repeated $m_1 = m_2$: $y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$.

Complex $m = \alpha \pm \beta i$: $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$.

n th-order, repeated roots: a real root m repeated k times contributes $e^{mx}, x e^{mx}, \dots, x^{k-1} e^{mx}$; a complex pair $\alpha \pm \beta i$ repeated k times contributes $e^{\alpha x} x^j \cos \beta x$ and $e^{\alpha x} x^j \sin \beta x$ for $j = 0, \dots, k-1$.

Undetermined coefficients – trial y_p forms: $g(x) = k$ (const) $\rightarrow A$; degree- n polynomial $\rightarrow A_n x^n + \dots + A_1 x + A_0$; $k e^{ax} \rightarrow A e^{ax}$; $k \cos \beta x$ or $k \sin \beta x \rightarrow A \cos \beta x + B \sin \beta x$ (always include both); product $\rightarrow$ product of matching forms (e.g. $x e^{ax} \cos \beta x$). If any assumed term duplicates a term in y_c , multiply the whole y_p guess by x (or x^2) until the duplication is gone.

Variation of parameters (2nd order): write $y'' + Py' + Qy = f(x)$ in standard form, $y_c = c_1 y_1 + c_2 y_2$, Wronskian $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$, $u_1' = \frac{-y_2 f(x)}{W}$, $u_2' = \frac{y_1 f(x)}{W}$, $y_p = u_1 y_1 + u_2 y_2$.

Cauchy-Euler equation (equidimensional), $x > 0$: $ax^2 y'' + bxy' + cy = 0$, sub $y = x^m \Rightarrow am(m-1) + bm + c = 0$. Distinct real: $y = c_1 x^{m_1} + c_2 x^{m_2}$; repeated m_1 : $y = c_1 x^{m_1} + c_2 x^{m_1} \ln x$; complex $\alpha \pm \beta i$: $y = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$.

General solution of a non-homogeneous linear DE: $y = y_c + y_p$ – y_c (complementary function) solves the associated homogeneous equation; y_p (particular solution) is any one solution of the full equation.

Laplace Transforms (Zill 4.1–4.5)

Definition: $\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$, $t \geq 0$ – converges for s large enough provided f is piecewise continuous and of exponential order. Linearity: $\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$.

Laplace transform pairs:

$$\mathcal{L}\{1\} = 1/s, s > 0, \quad \mathcal{L}\{t\} = 1/s^2, s > 0$$

$$\mathcal{L}\{t^n\} = n!/s^{n+1}, n = 1, 2, 3, \dots, s > 0$$

$$\mathcal{L}\{e^{at}\} = 1/(s-a), s > a$$

$$\mathcal{L}\{\sin kt\} = k/(s^2 + k^2), \quad \mathcal{L}\{\cos kt\} = s/(s^2 + k^2), s > 0$$

$$\mathcal{L}\{\sinh kt\} = k/(s^2 - k^2), \quad \mathcal{L}\{\cosh kt\} = s/(s^2 - k^2), s > |k|$$

$$\mathcal{L}\{t \sin kt\} = 2ks/(s^2 + k^2)^2, s > 0$$

$$\mathcal{L}\{t \cos kt\} = (s^2 - k^2)/(s^2 + k^2)^2, s > 0$$

$$\mathcal{L}\{e^{at} \sin kt\} = k/[(s-a)^2 + k^2], s > a$$

$$\mathcal{L}\{e^{at} \cos kt\} = (s-a)/[(s-a)^2 + k^2], s > a$$

$$\mathcal{L}\{e^{at} t^n\} = n!/(s-a)^{n+1}, s > a$$

$$\mathcal{L}\{\mathcal{U}(t-a)\} = e^{-as}/s, s > 0, a \geq 0$$
 – unit step

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}, a \geq 0$$
 – Dirac delta

Inverse Laplace transform: $\mathcal{L}^{-1}\{F(s)\} = f(t)$, linear: $\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\mathcal{L}^{-1}\{F(s)\} + b\mathcal{L}^{-1}\{G(s)\}$ – use partial fraction decomposition to match table entries, then invert term by term.

Transform of derivatives: $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$; $\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0)$; $\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - \dots - f^{(n-1)}(0)$ – turns linear DEs with initial conditions into algebraic equations in s .

Transform of an integral: $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$ – dividing by s corresponds to integrating f from 0 to t .

First translation theorem: $\mathcal{L}\{e^{at} f(t)\} = F(s-a)$ – multiplying $f(t)$ by e^{at} shifts the transform: replace every s in $F(s)$ by $(s-a)$.

Second translation theorem: $\mathcal{U}(t-a) = 0$ for $t < a$, 1 for $t \geq a$. $\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} = e^{-as} F(s)$; if $g(t)$ is not already a shift of a known function, use $\mathcal{L}\{g(t)\mathcal{U}(t-a)\} = e^{-as} \mathcal{L}\{g(t+a)\}$.

Transform of a periodic function: $\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$ – valid when $f(t+T) = f(t)$ for all $t \geq 0$; only one period needs to be integrated.

Dirac delta (unit impulse): $\int_0^\infty \delta(t-a) dt = 1$, $\int_0^\infty f(t) \delta(t-a) dt = f(a)$; $\mathcal{L}\{\delta(t-a)\} = e^{-as}$, $a \geq 0$; at $a = 0$, $\mathcal{L}\{\delta(t)\} = 1$.

Convolution theorem: $(f*g)(t) = \int_0^t f(\tau)g(t-\tau) d\tau$, and $\mathcal{L}\{f*g\} = F(s)G(s)$ – convolution in t is ordinary multiplication in s ; useful for inverting a product of two known transforms.

Vector Functions & Space Curves (Zill 9.1–9.3)

Vector-valued function & derivative: $\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$, $\vec{r}'(t) = f'(t)\hat{i} + g'(t)\hat{j} + h'(t)\hat{k}$ – $\vec{r}'(t)$ is tangent to the space curve where $\vec{r}'(t) \neq \vec{0}$.

Unit tangent vector: $\vec{T}(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$ – always unit length, points in the direction of increasing t .

Arc length: $L = \int_a^b \|\vec{r}'(t)\| dt = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt$.

Arc length function & speed: $s(t) = \int_a^t \|\vec{r}'(u)\| du$, $\frac{ds}{dt} = \|\vec{r}'(t)\| = v(t)$ – reparametrizing by arc length gives unit speed, $\|dr/ds\| = 1$.

Curvature: $\kappa = \frac{\|\vec{T}'(t)\|}{\|\vec{r}'(t)\|} = \frac{\|\vec{r}'(t) \times \vec{r}''(t)\|}{\|\vec{r}'(t)\|^3}$ – the cross-product

form is usually fastest by hand; large κ means sharp bending.

Principal normal & binormal: $\vec{N}(t) = \frac{\vec{T}'(t)}{\|\vec{T}'(t)\|}$, $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$
 $-T, N, B$ form a right-handed, orthogonal moving (Frenet) frame.

Frenet-Serret formulas: $\frac{d\vec{T}}{ds} = \kappa\vec{N}$, $\frac{d\vec{N}}{ds} = -\kappa\vec{T} + \tau\vec{B}$, $\frac{d\vec{B}}{ds} = -\tau\vec{N}$
 $-\tau$ (torsion) measures twisting out of the osculating plane.

Velocity, acceleration & tangential/normal components: $\vec{v}(t) = \vec{r}'(t)$, $\vec{a}(t) = \vec{r}''(t) = a_T\vec{T} + a_N\vec{N}$, $a_T = v'(t)$, $a_N = \kappa[v(t)]^2$ - a_T changes speed, a_N changes direction and depends on κ .

Scalar/Vector Fields, Gradient, Divergence, Curl (Zill 9.5-9.7)

$\phi(x, y, z)$ - scalar field, a single number at each point (temperature, pressure); $\vec{F}(x, y, z) = P\hat{i} + Q\hat{j} + R\hat{k}$ - vector field (velocity, force, flux).

Directional derivative: $D_{\vec{u}}f = \nabla f \cdot \vec{u}$, $\|\vec{u}\| = 1$ - instantaneous rate of change of f in direction $\vec{u}$.

Gradient vector: $\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$ - points toward max increase of f , magnitude = that max rate; ∇f is normal to the level surface $f = c$.

Divergence: $\text{div } \vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$ - scalar measuring net outward flux per unit volume; source where $\text{div } \vec{F} > 0$, sink where < 0 .

Curl: $\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix}$ - vector measuring

local rotation (circulation density); $\text{curl } \vec{F} = \vec{0}$ everywhere means F is irrotational.

Divergence/curl identities: $\nabla \times (\nabla f) = \vec{0}$ (curl of a gradient is always $\vec{0}$); $\nabla \cdot (\nabla \times \vec{F}) = 0$ (divergence of a curl is always 0) - explains why conservative fields are irrotational.

Laplacian operator: $\nabla^2 f = \nabla \cdot \nabla f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ - f is harmonic if $\nabla^2 f = 0$ throughout a region.

Line Integrals, Path Independence & Conservative Fields (Zill 9.8-9.9)

Line integral of a scalar field: $\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \|\vec{r}'(t)\| dt$ - mass of a wire, average value along C , etc.

Line integral of a vector field (work): $W = \int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt = \int_C P dx + Q dy + R dz$ - reversing the orientation of C negates the integral.

Fundamental theorem for line integrals: if $\vec{F} = \nabla f$ (F conservative, f a potential function), $\int_C \vec{F} \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$ - depends only on the endpoints of C , not the path (path independence).

Test for a conservative field: plane, $\vec{F} = P\hat{i} + Q\hat{j}$ conservative $\iff \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$; space, conservative $\iff \nabla \times \vec{F} = \vec{0}$ - requires F continuously differentiable on a simply connected region.

Finding a potential function: $f(x, y) = \int P(x, y) dx + g(y)$, then match $\frac{\partial f}{\partial y} = Q(x, y)$ to solve for $g(y)$ - same technique used for exact differential equations.

Closed-loop property: $\vec{F}$ conservative $\iff \oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed curve C - work done around any closed path in a conservative field is zero.

Green's Theorem, Surface Integrals & Flux (Zill 9.10-9.13)

Green's theorem: let C be a positively oriented, piecewise-smooth, simple closed curve bounding a region D , with P, Q having continuous first partial derivatives on an open region containing D :

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

C is traversed counterclockwise.

Area via Green's theorem: $A = \oint_C x dy = -\oint_C y dx = \frac{1}{2} \oint_C (x dy - y dx)$ - any of the three forms gives the enclosed area as a boundary line integral.

Parametrized surfaces & surface area: $\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}$, $A(S) = \iint_D \|\vec{r}_u \times \vec{r}_v\| dA$ - $\vec{r}_u \times \vec{r}_v$ is normal to the surface, its magnitude is the local area-scaling factor.

Surface integral of a scalar field: $\iint_S f(x, y, z) dS = \iint_D f(\vec{r}(u, v)) \|\vec{r}_u \times \vec{r}_v\| dA$ - for $z = g(x, y)$: $dS = \sqrt{g_x^2 + g_y^2 + 1} dA$ over the shadow region.

Flux integral (surface integral of a vector field): $\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \hat{n} dS = \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) dA$ - net flow of F through S in the direction of the chosen unit normal $\hat{n}$.

Flux through $z = g(x, y)$, upward normal: $\iint_S \vec{F} \cdot d\vec{S} = \iint_D \left(-P \frac{\partial g}{\partial x} - Q \frac{\partial g}{\partial y} + R \right) dA$ - D is the projection of S onto the xy -plane; flip the sign for a downward normal.

Divergence Theorem, Stokes' Theorem & Coordinate Systems (Zill 9.14-9.16)

Divergence theorem (Gauss): let Q be a solid region bounded by a closed, piecewise-smooth surface S with outward unit normal $\hat{n}$, F with continuous first partial derivatives on Q :

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_Q \nabla \cdot \vec{F} dV$$

converts a flux (surface) integral into a triple (volume) integral.

Stokes' theorem: let S be an oriented, piecewise-smooth surface bounded by a simple, closed, piecewise-smooth curve C oriented consistently with $\hat{n}$, F with continuous first partial derivatives on an open region containing S :

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

converts a closed line integral (circulation) into a surface integral of the curl.

Green's theorem as a special case of Stokes' theorem: $\hat{n} = \hat{k}$, $S \subset xy$ -plane $\implies \oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{k} dA = \oint_C P dx + Q dy$ - Green's theorem is the flat, two-dimensional case of Stokes' theorem.

Cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, $dV = r dz dr d\theta$ - natural for regions with an axis of symmetry.

Spherical coordinates: $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$, $dV = \rho^2 \sin \phi d\rho d\phi d\theta$ - ρ = distance from origin, ϕ = polar angle from $+z$ -axis ($0 \leq \phi \leq \pi$), θ = azimuthal angle.

Double integral in polar coordinates: $\iint_D f(x, y) dA = \iint_D f(r \cos \theta, r \sin \theta) r dr d\theta$ – shortcut for regions that are discs, annuli, or sectors.

Quick Reference – The Four Fundamental Theorems of Vector Calculus

Fundamental theorem for line integrals ($\vec{F} = \nabla f$ conservative):
 $\int_C \nabla f \cdot d\vec{r} = f(\text{end}) - f(\text{start})$.

Green's theorem ($C = \partial D$, plane region D): $\oint_C P dx + Q dy = \iint_D (Q_x - P_y) dA$.

Stokes' theorem ($C = \partial S$, oriented surface S): $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\text{curl } \vec{F}) \cdot d\vec{S}$.

Divergence theorem ($S = \partial Q$, closed surface, solid Q): $\iiint_S \vec{F} \cdot d\vec{S} = \iiint_Q (\text{div } \vec{F}) dV$.

All four relate an integral over a boundary to an integral of a derivative-type quantity over the region it bounds – the same idea as the ordinary FTC, one dimension at a time.

Quick Reference – First-Order ODE Method by Type

Separable ($dy/dx = g(x)h(y)$) – separate variables, integrate both sides.

Linear ($y' + P(x)y = Q(x)$) – integrating factor $\mu = e^{\int P dx}$.

Exact ($M dx + N dy = 0$, $M_y = N_x$) – find potential f with $f_x = M$, $f_y = N$; solution $f(x, y) = C$.

Non-exact – multiply by an integrating factor $\mu(x)$ or $\mu(y)$ to restore $M_y = N_x$, then solve as exact.

Homogeneous ($dy/dx = f(y/x)$) – sub $y = vx$, reduces to separable equation in v .

Bernoulli ($y' + P y = Q y^n$) – sub $w = y^{1-n}$, reduces to linear equation in w .

Quick Reference – Root-Case Forms Side by Side

Constant-coefficient $ay'' + by' + cy = 0$ (auxiliary eqn. $am^2 + bm + c = 0$) vs. Cauchy-Euler $ax^2y'' + bxy' + cy = 0$ (auxiliary eqn. $am(m-1) + bm + c = 0$, sub $y = x^m$):

Distinct real m_1, m_2 – const. coeff.: $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$;
 Cauchy-Euler: $y = c_1 x^{m_1} + c_2 x^{m_2}$.

Repeated m_1 – const. coeff.: $y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$;
 Cauchy-Euler: $y = c_1 x^{m_1} + c_2 x^{m_1} \ln x$.

Complex $\alpha \pm \beta i$ – const. coeff.: $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$;
 Cauchy-Euler: $y = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$.

Quick Reference – Solving an IVP by Laplace Transform

1) Apply $\mathcal{L}\{\cdot\}$ to both sides; use $\mathcal{L}\{y^{(n)}\} = s^n Y(s) - s^{n-1}y(0) - \dots - y^{(n-1)}(0)$ to convert the DE into an algebraic equation in $Y(s) = \mathcal{L}\{y\}$.

2) Solve algebraically for $Y(s)$.

3) Decompose $Y(s)$ by partial fractions into pieces matching table entries; apply the shifting theorems for terms with e^{as} or e^{-as} factors, or the convolution theorem for a leftover product $F(s)G(s)$.

4) Invert term by term: $y(t) = \mathcal{L}^{-1}\{Y(s)\}$.

Quick Reference – Which Vector Calculus Theorem Applies

$\vec{F} = \nabla f$, want $\int_C \vec{F} \cdot d\vec{r}$ along an open curve $\rightarrow$ fundamental theorem for line integrals (endpoints only).

Closed curve C bounding a plane region D , want $\oint_C P dx + Q dy \rightarrow$ Green's theorem.

Closed curve C bounding an oriented surface S in space, want circulation $\oint_C \vec{F} \cdot d\vec{r} \rightarrow$ Stokes' theorem (reduces to Green's when S is flat).

Closed surface S bounding a solid Q , want flux $\iint_S \vec{F} \cdot d\vec{S} \rightarrow$ divergence theorem.

Quick Reference – Space-Curve Quantities at a Glance

$\vec{T} = \vec{r}' / \|\vec{r}'\|$ (unit tangent), $\vec{N} = \vec{T}' / \|\vec{T}'\|$ (principal normal), $\vec{B} = \vec{T} \times \vec{N}$ (binormal) – together the Frenet frame.

$\kappa = \|\vec{r}' \times \vec{r}''\| / \|\vec{r}'\|^3$ (curvature, cross-product form); τ from Frenet-Serret (torsion – twisting out of the osculating plane).

$L = \int_a^b \|\vec{r}'(t)\| dt$ (arc length); $v(t) = ds/dt = \|\vec{r}'(t)\|$ (speed).

$\vec{a} = a_T \vec{T} + a_N \vec{N}$, with $a_T = v'(t)$ (changes speed) and $a_N = \kappa v(t)^2$ (changes direction).

Quick Reference – Field Operators at a Glance

∇f – vector; points toward max increase of scalar field f ; normal to level surfaces of f .

$\nabla \cdot \vec{F}$ – scalar; net outward flux per unit volume (source if > 0 , sink if < 0).

$\nabla \times \vec{F}$ – vector; local rotation/circulation density; $\vec{0}$ everywhere means F is irrotational.

$\nabla^2 f = \nabla \cdot \nabla f$ – scalar; sum of unmixed second partials; f is harmonic where $\nabla^2 f = 0$.

$\nabla \times (\nabla f) = \vec{0}$ always, and $\nabla \cdot (\nabla \times \vec{F}) = 0$ always – gradients are irrotational, curls are divergence-free.

Quick Reference – Area/Volume Elements by Coordinate System

Cartesian: $dA = dx dy$, $dV = dx dy dz$.

Polar: $dA = r dr d\theta$, with $x = r \cos \theta$, $y = r \sin \theta$.

Cylindrical: $dV = r dz dr d\theta$, with $x = r \cos \theta$, $y = r \sin \theta$, $z = z$.

Spherical: $dV = \rho^2 \sin \phi d\rho d\phi d\theta$, with $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$.

Quick Reference – Exactness & Conservative-Field Tests Side by Side

Exact DE: $M dx + N dy = 0$ is exact $\iff M_y = N_x$; find potential f with $f_x = M$, $f_y = N$, solution $f(x, y) = C$.

Conservative field: plane $\vec{F} = P\hat{i} + Q\hat{j}$ conservative $\iff P_y = Q_x$;
 space $\iff \nabla \times \vec{F} = \vec{0}$; find potential f with $\nabla f = \vec{F}$.

Both use the same integrate-and-match procedure to build the potential function – exact differential equations and conservative vector fields are the same underlying idea.